

KALAELOA, HI, USA

WMO: 911780

Lat: 21.317N

Lon: 158.067W

Elev: 33

StdP: 14.68

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22551

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	60.7	62.6	51.2	56.0	67.7	53.2	60.2	68.8	23.0	75.5	20.2	76.3	3.5	40	0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.5	89.7	74.1	88.4	73.6	87.7	73.4	77.8	84.9	76.7	84.3	75.9	83.7	10.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	133.5	81.8	74.4	128.6	81.1	73.2	123.4	80.5	41.2	84.9	40.0	84.2	39.2	83.8	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.0	16.1	14.2	DB	54.9	91.3	6.4	2.4	50.3	93.0	46.5	94.4	42.9	95.8	38.3	97.5
			WB	52.4	79.6	3.0	3.1	50.2	81.8	48.5	83.6	46.8	85.3	44.6	87.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.6	72.6	72.7	73.4	75.1	76.4	78.7	80.0	80.5	80.0	78.7	76.4	74.1
	DBStd	3.60	2.68	2.48	2.36	2.29	2.24	1.78	1.74	1.60	1.96	2.24	2.24	2.43
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9692	699	634	724	752	819	861	930	947	899	888	792	747
	CDD65	4219	235	214	259	302	354	411	465	482	449	424	342	282
	CDH74	31467	1141	1053	1344	1834	2597	3453	4255	4577	4096	3458	2204	1455
	CDH80	8248	72	69	132	299	583	1051	1450	1604	1397	1017	435	139
(14) Wind	WSAvg	6.9	7.2	7.3	7.4	7.4	7.0	7.1	7.1	7.0	6.3	6.3	6.4	6.8
(15) Precipitation	PrecAvg	18.5	3.3	2.0	1.5	1.5	0.8	0.3	0.2	0.2	0.5	2.1	2.1	2.9
	PrecMax	33.0	10.9	8.0	6.6	12.0	4.4	1.7	0.6	1.7	2.7	13.5	11.7	12.2
	PrecMin	4.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1
	PrecStd	8.1	3.2	2.1	1.6	2.2	1.0	0.4	0.2	0.3	0.5	3.3	2.4	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.1	83.2	84.4	86.2	87.7	89.8	90.3	90.5	90.4	89.4	87.0	84.2
		MCWB	70.6	70.4	70.0	71.1	73.0	73.7	73.7	74.6	74.5	74.9	73.6	71.8
	2%	DB	81.7	81.8	82.4	84.3	85.9	87.9	88.9	89.4	89.1	87.8	85.1	82.5
		MCWB	70.3	70.0	69.6	70.5	71.8	72.7	73.5	74.1	73.9	74.0	73.0	71.0
	5%	DB	80.9	80.9	81.4	82.9	84.4	86.4	87.9	88.2	87.9	86.4	83.9	81.5
		MCWB	70.0	69.6	69.7	70.1	71.3	72.0	73.2	73.6	73.7	73.6	72.6	70.6
	10%	DB	79.5	79.5	80.5	81.7	83.1	85.1	86.4	87.1	86.4	85.0	82.4	80.7
		MCWB	69.4	69.0	69.3	69.7	71.0	71.5	72.7	73.3	73.4	73.3	72.0	70.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	75.0	74.8	75.4	76.1	76.4	78.0	79.6	78.5	79.0	78.0	75.9
		MCDB	79.8	80.0	79.3	81.7	84.1	85.2	85.8	86.3	85.6	84.9	83.5	80.5
	2%	WB	73.6	73.3	73.2	73.3	74.5	75.1	76.3	77.5	77.2	76.8	76.1	74.1
		MCDB	78.3	78.6	78.9	81.0	82.7	84.4	84.7	85.3	84.9	84.3	82.0	79.4
	5%	WB	72.5	71.9	71.9	72.2	73.3	73.8	75.2	76.2	76.1	75.7	74.9	73.0
		MCDB	77.9	78.1	78.6	80.2	81.7	83.3	84.3	84.6	84.2	83.5	81.0	78.8
	10%	WB	71.0	70.5	70.6	70.9	72.3	72.9	74.1	75.0	75.1	74.8	73.4	71.6
		MCDB	77.3	77.4	77.9	79.4	80.9	82.6	83.4	83.7	83.5	82.5	80.0	78.1
(35) Mean Daily Temperature Range	MDBR		14.6	14.4	14.2	14.3	14.5	14.6	14.5	14.5	14.6	14.1	13.2	13.5
	5% DB	MCDBR	15.0	15.1	15.1	15.1	15.3	15.4	15.2	15.2	15.3	14.9	14.0	14.0
		MCWBR	7.9	7.7	7.4	6.7	6.6	5.9	5.8	6.0	6.0	6.2	6.7	6.8
	5% WB	MCDBR	12.7	13.1	13.8	14.4	14.4	14.6	14.3	14.1	14.0	13.9	13.0	12.7
		MCWBR	8.0	7.9	8.1	7.8	7.6	7.0	6.9	7.1	7.0	7.3	7.8	7.3
(40) Clear-Sky Solar Irradiance	taub		0.358	0.365	0.373	0.390	0.383	0.371	0.369	0.368	0.361	0.373	0.371	0.354
	taud		2.591	2.562	2.548	2.481	2.532	2.573	2.589	2.591	2.615	2.592	2.589	2.625
	Ebn at Noon		291	295	296	290	289	290	290	292	295	289	284	289
	Edh at Noon		29	31	33	36	34	32	31	31	30	30	29	27
(44) All-Sky Solar Radiation	RadAvg		1418	1594	1827	2061	2203	2301	2269	2181	2038	1715	1448	1310
	RadStd		62	114	193	118	128	66	47	87	61	56	39	94

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+122

Nomenclature: See separate page